

PAPERMIND COMPREHENSIVE EXAMINATION - COMPUTER SCIENCE

Time Allowed: 3 Hours

Maximum Marks: 100

1. What is the primary purpose of Database Normalization?
 - (A) To eliminate redundant data and avoid update anomalies
 - (B) To maximize storage usage on hard drives
 - (C) To increase SQL query parsing latency
 - (D) To automatically encrypt all stored data

2. Which data structure operates under the Last-In, First-Out (LIFO) principle?
 - (A) Queue
 - (B) Stack
 - (C) Binary Search Tree
 - (D) Hash Map

3. What is the time complexity of searching an element in a balanced Binary Search Tree?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n \log n)$

4. In Python, which keyword is used to create an anonymous function?

- (A) `def`
- (B) `lambda`
- (C) `func`
- (D) `anonymous`